

Tris(oxalato) Teaching Aids

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The preparation and characterization of potassium tris(oxalato)ferrate(III) is an experiment of recognized value to undergraduate chemistry courses (1, 2). We have enhanced the teaching potential of such compounds in a more challenging experiment based on the readily available members of the series $K_3[M(ox)_3] \cdot xH_2O$ (ox = oxalato; for M = Cr, Fe, Al, $x = 3$; for M = Co, $x = 3.5$). In this experiment, their preparation and comparative study demonstrates a number of synthetic strategies and valuable, although often neglected, instrumental techniques. The experiment is aimed at relatively small groups of capable students, and the various aspects may be selectively combined according to the available time and equipment. The experiment requires several half-day laboratory periods.

The Experiment

Preparation and Analysis of Compounds

The compounds are readily obtained in high yield, purity, and crystallinity by procedures incorporating a variety of preparative techniques and strategies (3). Capable students are able to prepare several compounds simultaneously. Methods for the chemical analysis of the compounds are also available (3, 4).

Infrared Spectra (as Nujol Mulls)

The infrared spectra of all the compounds are virtually identical, displaying the characteristic bands of the bidentate oxalato ligand. The IR spectra support the presence of isostructural tris(oxalato)metallate(III) anions in all compounds, this unit being maintained despite differing metal ions, formulations (differing x) and crystal structures (see below). Assignment for the observed IR bands may be found in the literature (5).

X-Ray Powder Diffractometry

Although an isostructural $[M(ox)_3]^{3-}$ anion is common to all the compounds, they are not all crystallographically isomorphous. The unit cells of the compounds, all of which belong to the monoclinic system, fall into two classes; very similar parameters pertain to the chromium, iron, and aluminum compounds whereas those of the cobalt compound are distinctly different (6). The extra water of crystallization in the cobalt compound leads to the larger unit cell and increased unit cell contents observed in this case, reflections of the unique crystal packing that accommodates these extra water molecules. This structural difference is clearly revealed by X-ray powder diffractometry, the powder pattern of $K_3[Co(ox)_3] \cdot 3.5H_2O$ being unique in comparison to the virtually identical patterns of the isomorphous $K_3[M(ox)_3] \cdot 3H_2O$ (M = Cr, Fe, Al) group. Powder patterns were recorded using an Automatic Philips P.W.1049 Powder Diffractometer employing Ni-filtered $Cu-K\alpha$ radiation ($\lambda = 1.5405 \text{ \AA}$); strong signals were readily obtained. The experimental patterns may also be compared with the theoretical patterns calculated from the unit cell dimensions (computer programs for the calcu-

lation of 2θ values are available in many modern crystallographic laboratories). Further, experimental peaks that are unmatched by a theoretical peak may be assigned to impurities, and these impurities (often oxalic acid or potassium oxalate) then may be identified through the American Society for Testing and Materials' "Power Diffraction Data File."¹ Alternatively, suspected impurities may be confirmed by examining the powder patterns of authentic samples (of the impurity).

Magnetic Susceptibility Measurements

The magnetic moment is an intrinsic property of any coordination compound and reflects among other things the metal's oxidation state, coordination number and geometry, and the nature of intermetallic interactions. The method we used to determine the solution magnetic susceptibilities of the title compounds features an invaluable technique to which many undergraduate students have little exposure—the Evans nuclear magnetic resonance (NMR) method (7, 8). Briefly, we employed coaxial NMR tubes, a Jeolco Minimar 100-MHz NMR spectrometer and solutions of the compounds in aqueous *tert*-butyl alcohol (2% in *t*-BuOH, $\chi_0 \sim -0.72 \times 10^{-6} \text{ g}^{-1}$). The spectra obtained exhibited two features of note: (1) paramagnetic broadening of the signal due to the compound-containing solution and (2) large spinning side bands. As a result, it was necessary to record a set of spectra at several tube spinning rates to distinguish the (invariant) resonance signals from the spinning side bands (which shift upon spinner adjustment). At 25°C, our typical results were: For M = Cr, concentration = 0.025 g/mL, $\Delta f = 68.8 \text{ Hz}$ giving $\mu_{\text{eff}} = 3.87 \text{ B.M.}$; for M = Fe, concentration = 0.008 g/mL, $\Delta f = 49.7 \text{ Hz}$ giving $\mu_{\text{eff}} = 5.98 \text{ B.M.}$; for M = Co, Al (conc. = 0.04 g/mL), diamagnetic. These magnetic moments are in very good agreement with the spin-only values for the respective metal ions, viz., Cr(III) d^3 , Fe(III) high-spin d^5 , Co(III) low-spin d^6 , and Al(III) d^{10} .

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A contribution from the Chemistry Department, The Faculties, Australian National University, Canberra, Australia.

¹ ASTM Publications are available from the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.